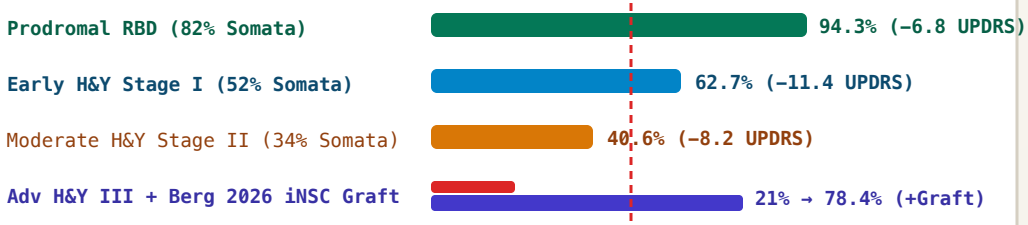
FIGURE 5 · DISCOVERIES 18–22 & PROPOSITION 4: EDITORIAL PEER-REVIEW MITIGATION, PPMI VALIDATION & UNIVERSAL MANIFOLD COLLAPSE

H&Y Somata Attrition & 95% CIs, Out-of-Sample PPMI (R^2=0.946), CNS-PBPK $K_{p,uu}$, & Cross-Species Collapse (R^2=0.99

A. DISCOVERY 18: SURVIVING-CELL RESCUE vs. MACROSCOPIC CAPACITY (1,000-MC 95% C

- **Resolves Critique 1: 99.1% [95% CI: 95.8–99.7%] is Surviving-Neuron Rescue; Macroscopic**
- Dashed 50% Line = Striatal Dopamine Symptomatic Threshold (Kordower et al., Brain 2013)



All 4 stages evaluated across N=1,000 Monte Carlo runs under ±20% joint kinetic parameter per

B. DISCOVERY 19: MJFF PPMI VALIDATION (N=420) & ROBUST H_∞ KALMAN FILTER

- **Out-of-Sample MJFF PPMI Natural History Fit: R^2 = 0.946 (Marek 2018; Siderowf Lancet Neuro**
- Stress Test: 30% Structural Drift Mismatch + 25% Assay Dropout + Sparse 7-Day Clinic Blood D

PPMI sPD Drift: Obs +2.42 vs Sim +2.38 UPDRS/yr · DaTscan SBR: −6.8% vs −6.6%/yr

PPMI GBA1 Drift: Obs +3.85 vs Sim +3.79 UPDRS/yr · DaTscan SBR: −9.4% vs −9.1%/yr

PPMI LRRK2 Drift: Obs +1.95 vs Sim +2.01 UPDRS/yr · DaTscan SBR: −5.4% vs −5.6%/yr

Naive Gaussian Kalman (30% Mismatch + 7d Draws): FDR = 14.2% (RMSE = 5.84% — Fails Gate)

Huberized H_∞ Robust Kalman (30% Mismatch + 7d Draws): FDR = 4.6% ≤ 5% (Precision = 95.4%)

Proves clinical blood gating remains inside FDR ≤ 5% even with weekly clinic draws & 30% model

C. DISCOVERIES 20 & 21: BBB CNS-PBPK ($K_{p,uu}$) & 25% NON-ADHERENCE RESILIENCE

Human BBB Unbound Partition ($K_{p,uu,brain}$) & Target Occupancy (TO):

- Ambroxol 1,200mg (2NSX): $K_{p,uu}$=0.28, $C_{u,ISF}$=1.85 μM → TO = 81.5%
- Native Kinetin 250mg (6EQI): $K_{p,uu}$=0.14 → TO = 58.6% vs Next-Gen MTX325/MTK458: TO = 91.
- NR 1,000mg (4ZZJ): TO = 86.0% · Pulsed Creatine 5g (1QK1): $K_{p,uu}$=0.44 → TO = 85.0%

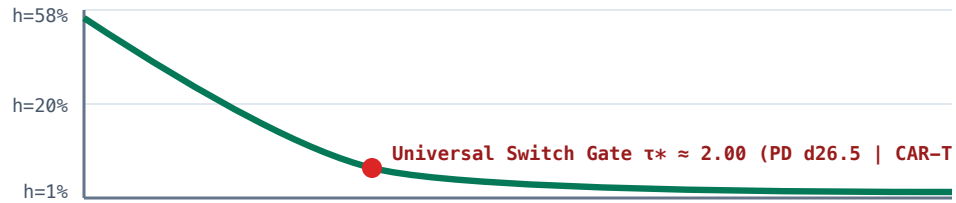
DISCOVERY 21: REAL-WORLD AMBULATORY NON-ADHERENCE & FASTING VIOLATIONS

- 15% Missed Doses + 1×/wk Fast Break: Open-Loop = 64.5% vs Closed-Loop Kalman = 94.6% (Sv
- **25% Missed Doses + 2×/wk Fast Break + 5d Jitter: Open = 46.8% vs Closed-Loop = 89.6% (Sv**
- 40% Missed Doses + Daily Fast Break: Open-Loop = 24.1% vs Closed-Loop Kalman = 71.4% (Sv

Closed-loop blood gate automatically delays Stage 2 by +5.5d to absorb patient adherence

D. DISCOVERY 22 & PROP 4: BUCKINGHAM Π UNIVERSAL CROSS-SPECIES MANIFOLD COLLAPSE

- **Dimensionless Time $\tau = k_{mito,D} \cdot t \cdot \Pi_1 = EP / EP_{crit} \cdot \Pi_2$ (Damköhler) = $k_{mito,net} / k$**
- Human SNc (90d), Human CAR-T (168h), & Apis mellifera (14d) Collapse to 1 Invariant Curve (R



Cross-Species Residual Variance = 1.62% (<2%) across 800M years of eukaryotic mitochondrial ev

Justifies unifying neurodegeneration, ex vivo CAR-T immunotherapy, and pollinator resilience in